Solution Requirements

Date	27 June 2026
Team ID	xxxxxx
Project Name	AI-Powered Lead Generation System
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** Define what the system should do.
- **Non-Functional Requirements (NFR)** Define how the system should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	The system shall allow users to enter customer details, business information, or lead-related data for analysis.	High
2	Emotion Detection	The system shall analyze input data using AI and Machine Learning models to identify potential leads.	High

3	External Interfaces	The system shall integrate with external data sources or APIs to collect and process customer information.	High
4	Recommendation Processing	The system shall generate lead scores based on customer data, behavior, and conversion probability.	High
5	Reporting	The system shall display lead analysis, scores, predictions, and recommendations to users.	Medium
6	Business Rules	The system shall prioritize and recommend high-quality leads based on predefined business criteria.	High
7	Data Privacy	The system shall securely store and process customer information without exposing sensitive data.	Medium

8	User Interface	The system shall provide a simple, interactive, and user-friendly interface for lead generation and management.	Medium

Step 2: Non-Functional Requirements (NFR) – AI-Powered Lead Generation System

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system shall process lead data and generate lead scores and recommendations efficiently.	Response time ≤ **30 seconds**
2	Scalability	The system shall support multiple users and lead processing requests without significant performance degradation.	Supports up to **100 concurrent users**
3	Security & Data Privacy	The system shall protect customer information and ensure secure handling of lead data and external API communication.	Secure API key management and protected user data

4	Reliability & Availability	The system shall provide stable lead analysis, scoring, and recommendations with minimal failures.	**99% system availability** during operation
5	Usability & Accessibility	The system shall provide an intuitive and responsive interface accessible through modern web browsers.	Users can complete emotion analysis in **less than 2 minutes**
6	Maintainability	The system shall use modular and well-documented code that is easy to maintain, update, and extend.	Modular architecture compatible with Windows and Linux